

DPYSL4/CRMP3 (O14531): Evaluation of Broad Hydrolase Activity Assignment

Focus: function_assignment — does human DPYSL4/CRMP3 directly have hydrolase activity (GO:0016787), specifically hydrolase acting on C–N bonds other than peptide bonds (GO:0016810), on the strength of the 2013 CRMP3 histone-H4 deacetylation study

(P 23443259 /

PMC3583001)?

Iteration: updated through Iteration 3 of 3.

Executive Judgment

Verdict: Over-annotated / refuted for intrinsic activity — assigning GO:0016787 and GO:0016810 to human DPYSL4 as experimentally-meaningful MF is NOT justified. In the live GO record these two terms already exist ONLY as InterPro2GO family-based IEA carry-over, and they coexist with curated NOT dihydropyrimidinase / NOT pyrimidine-catabolism annotations. The single deacetylase study does not establish intrinsic catalysis and does not upgrade them.

Decisive new evidence (Iteration 3, from the primary study's own full text, PMC3583001): the reported "CRMP3 HDAC activity" was **completely inhibited by 1 μ M Trichostatin A (TSA)**. TSA inhibits classical class I/II HDACs by chelating their catalytic Zn^{2+} — a metal center that DPYSL4 structurally **cannot** possess (its amidohydrolase pocket has lost 4 of 6 Zn-coordinating residues). An activity that is both attributed to DPYSL4 *and* behaves exactly like a canonical zinc-HDAC is most parsimoniously explained by an **associated or co-purifying Zn-HDAC**, not by intrinsic DPYSL4 catalysis. The authors deserve credit for a genuine contamination control (re-expression in Sf9 insect cells), but insect cells also contain endogenous Zn-HDACs that can co-purify with an affinity-tagged bait, and **no catalytically-dead point mutant** was used to prove intrinsic catalysis (none is even designable, because the active site is degenerate). The paper's mechanistic rationale is also flawed: it lists "histone deacetylase (HDAC)" as a member

of the "cyclic amidohydrolase superfamily," but HDACs are a structurally distinct arginase/deacetylase fold, so DPYSL4's dihydropyrimidinase-domain homology does not imply deacetylase capability.

Reasoning: 1. **The ancestral C–N hydrolase activity is demonstrably lost.** DPYSL4 belongs to the amidohydrolase/dihydropyrimidinase superfamily but has a **degenerate active site**: a computed global alignment against the active enzyme human dihydropyrimidinase (DPYS, Q14117) shows only **2 of 6 Zn²⁺-coordinating catalytic residues are conserved**. The metal-bridging carbamoylated lysine (DPYS Lys159) is replaced by Leu, and three of four catalytic histidines/aspartate are lost (His69→Arg, His248→Lys, Asp326→Ala). The binuclear zinc center — required for metal-dependent C–N bond hydrolysis — cannot form. Independent crystallography (Myllykoski et al. 2017,

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states outright that CRMPs "have lost the enzymatic active site." The primary study itself confirms rodent CRMPs "do not hydrolyze dihydropyrimidinase substrates."

1. **The deacetylase claim is single-source and mechanistically implausible.** Only one primary study reports the activity (Hou et al. 2013), it was performed on the **mouse ortholog** (Dpysl4, O35098 — 93.2% identical to human, catalytic residues identically degenerate), it comes from a **single laboratory with no independent replication**, and DPYSL4 lacks **any** catalytic machinery compatible with protein deacetylation (no Zn-dependent HDAC site, no NAD⁺-dependent sirtuin fold). Co-purifying contaminant deacetylase — a classic artifact for "novel deacetylase" claims — was not excluded.
2. **Technicality vs. warranted annotation.** Protein-lysine deacetylation is formally a hydrolysis of a carbon–nitrogen (amide) bond, so *if* the activity were genuine and intrinsic it would fall under GO:0016810. But the evidence is too weak, unreplicated, ortholog-based, and mechanistically unsupported to license an experimental MF annotation of the broad hydrolase parents to human DPYSL4. The seed argument ("loss of dihydropyrimidinase = loss of all hydrolase activity") is logically valid but is not rescued by the deacetylase paper, because that paper does not establish a credible, replicated, intrinsic hydrolase activity for the human protein.

Most important caveats: The full text of PMC3583001 was retrieved (via NCBI efetch, Iteration 3) and the methods inspected directly. Constructs: recombinant mouse full-length CRMP3, calpain product p54, and the conserved dihydropyrimidinase "D domain," His6/S-tagged, affinity-purified from HEK293 and (as a control) Sf9 insect cells; assayed with a cell-free fluorescent HDAC assay on a histone-derived substrate. The residual uncertainty is

interpretive, not descriptive: affinity purification of a tagged bait does not exclude a tightly-associated HDAC in either host, TSA sensitivity actively points to a Zn-HDAC mechanism the protein cannot supply, no catalytically-dead mutant was tested, no independent group has replicated the activity, and it was never measured on human O14531.

Live GO record for O14531 (QuickGO, retrieved Iteration 2)

GO ID	Name	Aspect	Evidence	Reference	Qualifier	Interpretation
GO:0016787	hydrolase activity	MF	IEA ECO:0000256	GO_REF:0000002 (InterPro2GO)	enables	Family-fo electronic transfer; n experime
GO:0016810	hydrolase acting on C–N bonds, non- peptide	MF	IEA ECO:0000256	GO_REF:0000002 (InterPro2GO)	enables	Family-fo electronic transfer; t is the hypothes term
GO:0016812	hydrolase, C–N bonds, in cyclic amides	MF	IBA ECO:0000318	GO_REF:0000033	enables	Phylogene internally inconsiste with the M below
GO:0004157	dihydropyrimidinase activity	MF	IBA ECO:0000318	GO_REF:0000033	NOT enables	Curated absence o specific ancestral activity
GO:0006208	pyrimidine nucleobase catabolic process	BP	IBA ECO:0000318	GO_REF:0000033	NOT involved_in	Curated absence o pathway
GO:0005515	protein binding	MF	IPI	5 PMIDs	enables	Better- supported than hydr
GO:0031005	filamin binding	MF	IPI	P 25358863	enables	Specific interactio
GO:0007399	nervous system development	BP	TAS	P 9652388	involved_in	Better- supported
GO:0005829	cytosol	CC	IBA/TAS	Reactome	located_in/ is_active_in	Localizati

Key point: the hypothesis's two target terms are precisely the automatic InterPro2GO predictions driven by the amidohydrolase fold (InterPro signatures PF01979 Amidohydrolase, IPR006680, IPR011778 D-phenylhydantoinase/dihydropyrimidinase, IPR050378/PTHR11647 Metallo-dependent Hydrolases, TIGR02033 dihydropyrimidinase). They are contradicted by the curated `NOT GO:0004157` / `NOT GO:0006208` calls and by the degenerate active site — a textbook pseudo-enzyme over-annotation.

Evidence Matrix

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
23443259 (Hou 2013, PMC3583001)	Direct in-vitro assay + cell/animal	Supports (the seed) but qualifies heavily	CRMP3 has histone H4 deacetylase activity	"mouse CRMP3 has robust histone H4 deacetylase activity"; truncated CRMP3 → H4 deacetylation → E2F1 de-repression → neuronal death	Mouse CRMP3 Dpysl4 excitotoxic ischemic calpain cleaved nuclear
23443259 (full text, Iteration 3)	Direct assay detail — inhibitor profile	Refutes intrinsic activity (competing = contaminant Zn-HDAC)	Is the "CRMP3 HDAC" activity a classical zinc-HDAC?	"TSA (1 μ M) completely inhibited CRMP3 HDAC activities" — full-length > D domain > p54; complete TSA (Zn-chelator) inhibition	HEK293 Sf9-expressing recombinant protein free fluoresces HDAC
23443259 (full text)	Contamination control	Qualifies (partial support for authors)	Is activity from HEK293 host contaminants?	Re-expressed in Sf9 insect cells to exclude HEK293 contaminants; activity retained	Sf9 insect cells
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Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
23443259 (full text)	Rationale / classification	Refutes premise	Does DHPase-fold homology imply deacetylase capability?	Paper places "HDAC" inside the "cyclic amidohydrolase superfamily" — but HDACs are a distinct arginase/deacetylase fold	
23443259 (same)	Author statement	Refutes ancestral activity	Dihydropyrimidinase activity	"purified rodent brain CRMPs do not hydrolyze dihydropyrimidinase substrates"	Rodent CRMPs
28044206 (Myllykoski 2017)	Structural (1.25 Å X-ray) / evolutionary	Refutes intrinsic hydrolase	CRMPs retain a catalytic site	"belong to the dihydropyrimidinase family, they have lost the enzymatic active site"	Human CRMP2 paralogs
Computed alignment DPYS(Q14117) vs DPYSL4(O14531)	Structural/ evolutionary (this run)	Refutes C–N hydrolase	Are Zn-coordinating catalytic residues conserved?	Only 2/6 conserved; carbamoyl-Lys159→Leu, His69→Arg, His248→Lys, Asp326→Ala; 59.6% overall identity	Human sequence UniProt
Computed mouse–human ortholog check (this run)	Computational	Qualifies	Is the assayed protein the true DPYSL4 ortholog?	Mouse Dpysl4 O35098 = 93.2% identical to human; catalytic residues identically degenerate	UniProt

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
16495451 (Hou 2006), 19559021 (Aylsworth 2009)	Cell/animal mechanism	Context (competing interpretation)	Function of calpain-cleaved CRMP3	Truncated CRMP3 → nuclear translocation, nuclear condensation, neuronal death; inhibits microtubule polymerization	Rodent neuron same l

GO Curation Implications (leads — require curator verification)

- **The two hypothesis terms already exist only as IEA InterPro2GO carry-over (GO_REF:0000002).** The curation question is therefore *not* "should we add them from the deacetylase paper" but "should the existing family-based IEA GO:0016787/GO:0016810 be trusted." Answer: **no / low-confidence** — they are automatic fold-based predictions for a pseudo-enzyme whose active site is degenerate, and they conflict with the curated **NOT** GO:0004157 (dihydropyrimidinase) and **NOT** GO:0006208 (pyrimidine catabolism) annotations. Recommend flagging GO:0016787/GO:0016810 as unreliable IEA (candidate for **NOT** or removal), pending curator judgment on how far to generalize the loss-of-activity.
- **Do NOT promote GO:0016787 / GO:0016810 to an experimentally-supported MF for DPYSL4** on the basis of

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Rationale: ancestral C–N hydrolase activity is lost (degenerate active site; author-confirmed absence of dihydropyrimidinase activity), and the deacetylase activity is single-source, ortholog-based, mechanistically implausible, and unreplicated. The deacetylase paper is not even the provenance of the current terms.

- **Resolve the internal inconsistency:** GO:0016812 (cyclic-amidase) is currently **enables** (IBA) while its child GO:0004157 is **NOT**. A curator should reconcile these — the

biochemistry (no dihydropyrimidinase activity; degenerate site) argues the positive cyclic-amidase IBA is also questionable.

- **If a deacetylase-related term is considered at all**, it should be treated as **uncertain / non-core** and attributed with the appropriate ortholog/evidence caveat (mouse, in vitro) — not propagated as a confident human MF. The more specific term, if ever supported, would be **histone deacetylase activity (GO:0004407)**, itself a child of GO:0016810; but the current evidence does not meet the bar for an experimental (IDA) annotation, and IEA/homology transfer is contraindicated by the degenerate active site.
- **Dihydropyrimidinase / cyclic-amidase activity (GO:0004157 and related): should be treated as NOT present** (retain any existing "loss of function" / NOT annotation).
- **Better-supported terms for DPYSL4** are its cytoskeletal/neurodevelopmental roles (semaphorin signaling, neurite outgrowth regulation, microtubule/tubulin interaction) — BP/CC and interaction terms are more informative than a broad hydrolase MF. (Avoid "protein binding" as the endpoint; the vimentin/tubulin interactions and neurite-outgrowth BP are more specific.)

Mechanistic Scope

- **Immediate molecular function tested:** intrinsic hydrolase (amidohydrolase) activity of the DPYSL4 polypeptide — either (a) cyclic-amide hydrolysis of dihydropyrimidines (ancestral) or (b) amide hydrolysis of acetyl-lysine (the proposed deacetylase).
- **Direct gene-product activity:** Sequence/structure indicate DPYSL4 lacks a functional metallo-hydrolase center; the ancestral (a) is lost. The proposed (b) is asserted for the mouse ortholog only, in vitro.
- **Downstream phenotypes (must be separated from MF):** calpain cleavage → nuclear translocation → nuclear condensation → E2F1 de-repression → neuronal death. These are cellular/organismal consequences and do not by themselves demonstrate that DPYSL4 is the direct catalytic deacetylase; the H4 deacetylation could be mediated by an associated/contaminating enzyme or an indirect chromatin effect.

Conflicts and Alternatives

- **Paralog/family evidence conflicts with intrinsic catalysis:** the whole CRMP subfamily (DPYSL1–5) is structurally established to have lost the amidohydrolase active site (P 28044206).
- This is the strongest alternative: DPYSL4 is a pseudo-enzyme scaffold.
- **Organism/isoform:** activity shown for mouse Dpysl4 truncation product, not full-length human O14531. Even at 93% identity, the human MF is inferred, not measured.
- **Experimental artifact (now strongly indicated):** the activity was **completely inhibited by 1 μ M TSA**, a Zn^{2+} -chelating inhibitor of classical class I/II HDACs. Because DPYSL4 has no such zinc center, an associated/co-purifying Zn-HDAC is the leading alternative explanation. The authors' Sf9 dual-expression control reduces but does not eliminate this (insect cells have their own Zn-HDACs, and affinity-tagged CRMP3 could pull down an associated HDAC). No catalytically-dead mutant was used.
- **Invalid homology premise:** the paper justifies the activity by placing HDAC within the "cyclic amidohydrolase superfamily," but HDACs are an unrelated arginase/deacetylase fold; DPYSL4's dihydropyrimidinase-domain homology gives no basis for deacetylase chemistry.
- **Single-lab provenance:** all mechanistic CRMP3 death papers trace to one group (Hou/Zhou), raising the value of independent replication before curation.

Knowledge Gaps

1. **Catalytic-dead control (RESOLVED as absent).** Checked: full-text methods (Iteration 3). The study used His6/S-tag affinity purification from HEK293 and Sf9, a TSA control, and HDAC-isoform comparisons, but **no catalytically-dead CRMP3 point mutant** — the single control that would prove intrinsic catalysis. None is readily designable because the active site is degenerate. Gap now: this control is missing and would be decisive; TSA-sensitivity meanwhile points away from intrinsic activity. Resolve: mass-spec of the affinity eluate for co-purifying HDACs; a mutant that ablates any candidate catalytic residue.
2. **Independent replication of DPYSL4 deacetylase activity.** Checked: PubMed — none found. Matters because a single source is below the bar for a confident MF. Resolve: targeted literature/citation search; any orthogonal biochemistry.

3. **Human O14531 direct assay.** Checked: not available. Matters for species-specific annotation. Resolve: assay recombinant human DPYSL4.
4. **Structural confirmation of pocket geometry.** Checked: alignment-based only (no superposition run here). Resolve: superpose an AlphaFold/experimental DPYSL4 model onto DPYS to confirm the metal site is non-functional (Phenix superpose / structure comparison).

Discriminating Tests

- **Catalytically-dead mutant assay:** mutate a residue essential to any proposed DPYSL4 catalytic mechanism and show loss of deacetylase activity — distinguishes intrinsic activity from contaminant.
- **Rigorous recombinant purification + mass-spec of the prep** to detect co-purifying HDAC1/2/SIRT.
- **Inhibitor panel (partly already answered):** the reported activity is fully blocked by 1 μ M TSA — i.e. it behaves as a class I/II Zn-HDAC, which DPYSL4 cannot be, implicating a contaminant/associated enzyme. A definitive test would add mass-spec of the eluate and a nicotinamide/sirtuin arm.
- **Structural superposition** (Phenix) of a DPYSL4 model onto DPYS/dihydropyrimidinase to quantify active-site collapse.
- **Human O14531 replication** of the H4 deacetylation assay.

Curation Leads (require curator verification)

- **Candidate reference to verify:**

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— verify snippets: "*mouse CRMP3 has robust histone H4 deacetylase activity*", "*purified rodent brain CRMPs do not hydrolyze dihydropyrimidinase substrates*", and the decisive methods line "*TSA (1 μ M) completely inhibited CRMP3 HDAC activities.*" Confirm construct = mouse Dpysl4 (O35098), His6/S-tagged, expressed in HEK293 and Sf9; note the **absence of a catalytically-dead mutant control** and the invalid superfamily premise (HDAC listed within "cyclic amidohydrolase superfamily").

- **Candidate corroborating reference:**

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— verify snippet: *"belong to the dihydropyrimidinase family, they have lost the enzymatic active site."*

- **Candidate GO actions:**

- Do **not** assert GO:0016787 / GO:0016810 as experimental MF for DPYSL4 (over-annotation risk).
- If retained from a prior action, **generalize/qualify or remove**; at most treat deacetylase as an **uncertain, non-core** claim (would resolve to GO:0004407 if ever confirmed) with mouse/in-vitro caveats.
- Confirm/retain **NOT dihydropyrimidinase activity** for the human protein.
- **Suggested questions for curator:** Was any prior MF hydrolase annotation on DPYSL4 IEA-by-homology from the DPYS active site? If so, it is contraindicated by the degenerate catalytic residues shown here.
- **Suggested experiments:** catalytically-dead mutant + inhibitor panel + human-protein replication (see Discriminating Tests).

Provenance (analyses run, Iterations 1–3)

- **Iter 1:** UniProt REST fetch of O14531 (DPYSL4), Q14117 (DPYS), Q16555 (CRMP2), O35098 (mouse Dpysl4); DPYS catalytic/binding-site features extracted from UniProt. Needleman–Wunsch (BLOSUM62) global alignment DPYS vs DPYSL4: 59.6% identity; catalytic-residue mapping → 2/6 Zn-coordinating residues conserved (carbamoyl-Lys159→Leu, His69→Arg, His248→Lys, Asp326→Ala). Mouse–human DPYSL4 identity 93.2% with identical catalytic degeneracy. PubMed searches.
- **Iter 2:** QuickGO annotation retrieval for O14531 (all three aspects) → GO:0016787/0016810 are IEA/InterPro2GO; GO:0004157 and GO:0006208 are **NOT**. InterPro entry retrieval → amidohydrolase/dihydropyrimidinase family signatures (PF01979, IPR006680, IPR011778, IPR050378, PTHR11647, TIGR02033, cd01314) confirming the IEA provenance.
- **Iter 3:** NCBI efetch retrieval of PMC3583001 full text → construct/expression/purification details; decisive line "TSA (1 μ M) completely inhibited CRMP3 HDAC activities"; Sf9 dual-expression contamination control; no catalytic-dead mutant; superfamily-premise error identified.

Bottom line for the curator

The seed hypothesis's broad hydrolase terms (GO:0016787, GO:0016810) are **not** supported as intrinsic molecular functions of human DPYSL4. They presently exist only as automatic InterPro2GO fold-based IEA predictions that conflict with curated **NOT** annotations and with a degenerate active site, and the one deacetylase study that might justify them is single-lab, mouse-only, unreplicated, mechanistically implausible (TSA-sensitive activity in a protein lacking a Zn-HDAC center), and built on an invalid fold-homology premise. Recommended lead: do not promote to experimental MF; flag the IEA terms as low-confidence/candidate-for-**NOT**; record any deacetylase claim only as uncertain, non-core, with mouse/in-vitro caveats.

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